

SEM Dendrite Segmentation: Comparative Technical Report

Images evaluated: 14
Failures (Dice<threshold): 8

Average metrics:
Classic Dice=0.845 IoU=0.787 Precision=0.788 Recall=0.932
YOLO Dice=0.449 IoU=0.302 Precision=0.335 Recall=0.709

This PDF was generated by fallback mode (matplotlib) because pdflatex was not available. The matching LaTeX source is summary_report.tex.

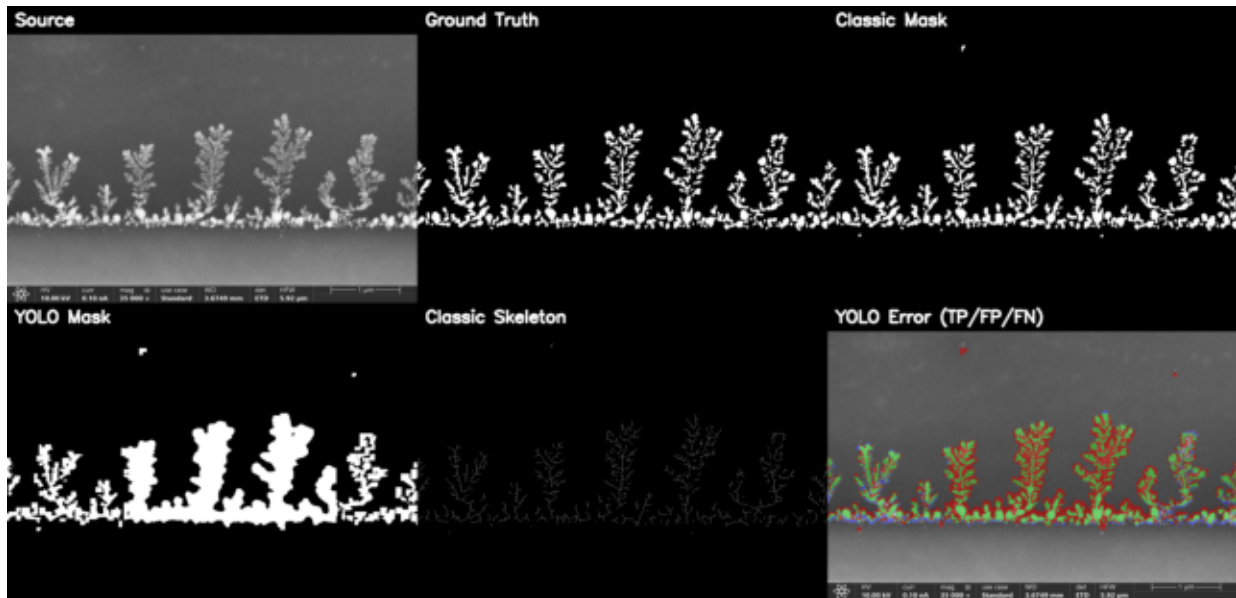
Per-image metrics

Image	C Dice	C IoU	C P	C R	Y Dice	Y IoU	Y P	Y R
2e-9_100s_00	0.923	0.857	0.857	1.000	0.579	0.407	0.409	0.991
2e-9_100s_01	0.931	0.872	0.872	1.000	0.655	0.487	0.560	0.789
2e-9_100s_01	0.699	0.538	0.538	1.000	0.103	0.055	0.066	0.241
2e-9_100s_01	0.997	0.995	0.995	1.000	0.596	0.424	0.454	0.865
diameter_100nm	0.927	0.863	0.863	1.000	0.438	0.280	0.302	0.798
diameter_100nm	0.919	0.850	0.850	1.000	0.523	0.354	0.399	0.758
diameter_100nm	0.995	0.991	0.991	1.000	0.454	0.293	0.374	0.577
diameter_100nm	1.000	1.000	1.000	1.000	0.359	0.218	0.272	0.527
diameter_100nm	0.024	0.012	0.017	0.043	0.065	0.034	0.039	0.194
Ag_1e-8_008	1.000	1.000	1.000	1.000	0.436	0.278	0.325	0.660
Ag_1e-9_02_0	0.906	0.828	0.828	1.000	0.560	0.389	0.432	0.796
Ag_2e-9_013	0.760	0.612	0.612	1.000	0.518	0.350	0.351	0.994
Ag_2e-9_014	0.753	0.603	0.603	1.000	0.479	0.315	0.316	0.994
Ag_40nm_pitch_	1.000	1.000	1.000	1.000	0.514	0.346	0.395	0.736

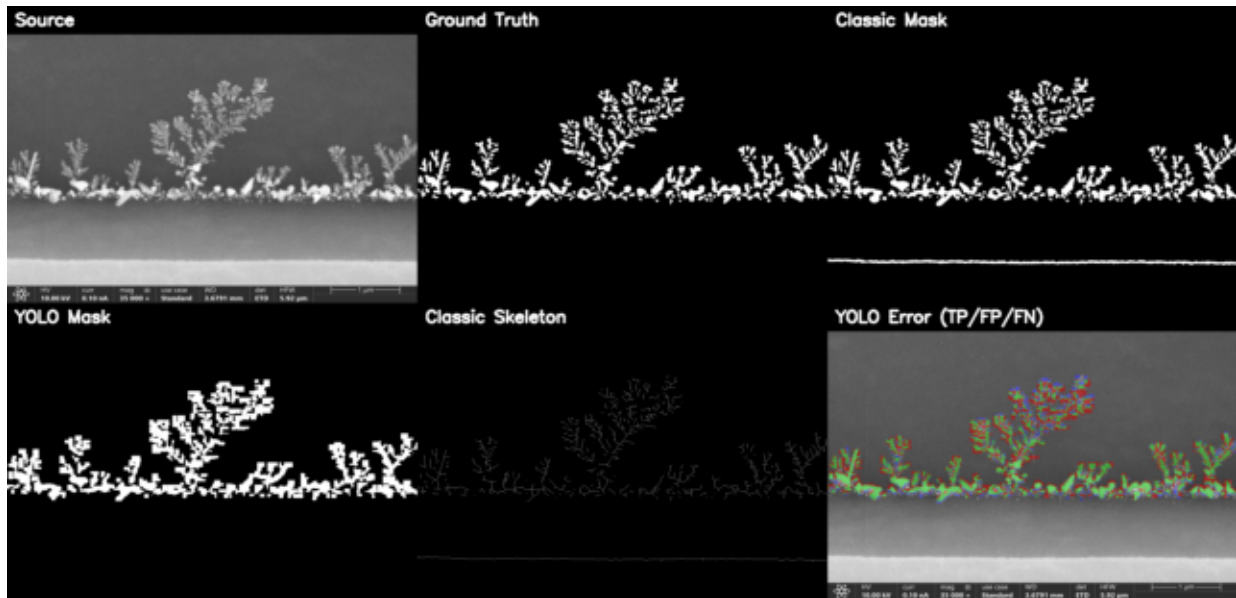
Failure analysis

Image	Method	Dice	Precision	Recall	Cause
0nm_diameter_100nm_pit	Classic	0.024	0.017	Fundamental mismatch - wrong region or severe an	
0nm_diameter_100nm_pit	YOLO	0.065	0.039	Fundamental mismatch - wrong region or severe an	
2e-9_100s_017	YOLO	0.103	0.066	Fundamental mismatch - wrong region or severe an	
0nm_diameter_100nm_pit	YOLO	0.359	0.272	Fundamental mismatch - wrong region or severe an	
Ag_1e-8_008	YOLO	0.436	0.325	Over-segmentation - noise included as dendrite (C	
0nm_diameter_100nm_pit	YOLO	0.438	0.302	Over-segmentation - noise included as dendrite (C	
0nm_diameter_100nm_pit	YOLO	0.454	0.374	Fundamental mismatch - wrong region or severe an	
Ag_2e-9_014	YOLO	0.479	0.316	Over-segmentation - noise included as dendrite (C	

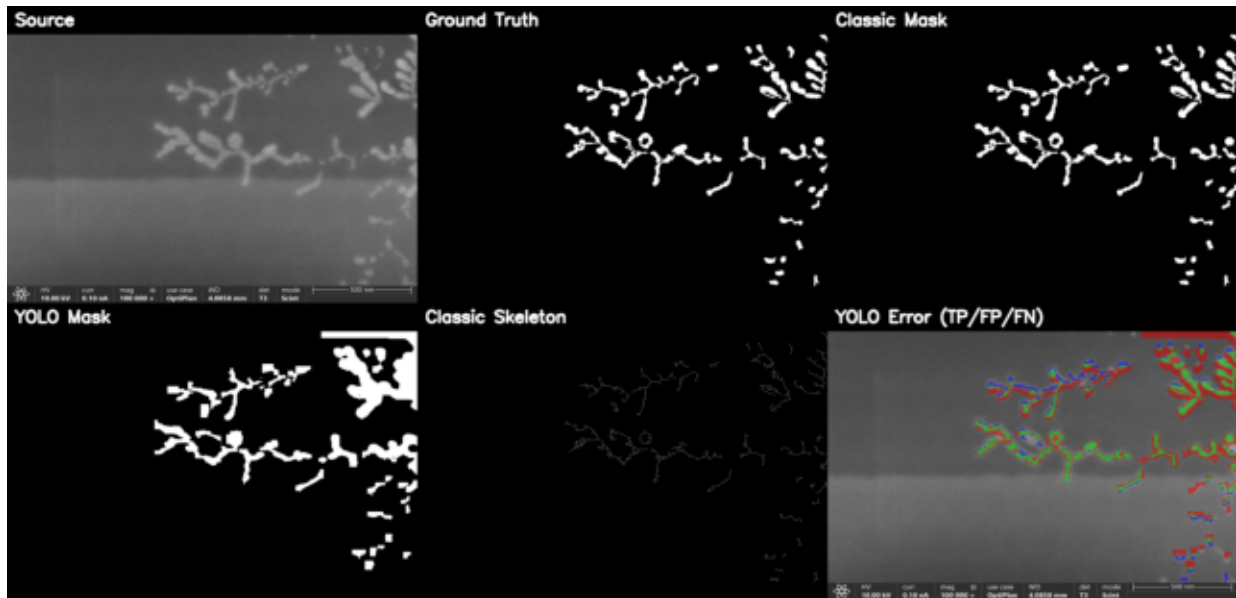
Success case: 2e-9_100s_019 (combined Dice=0.796)



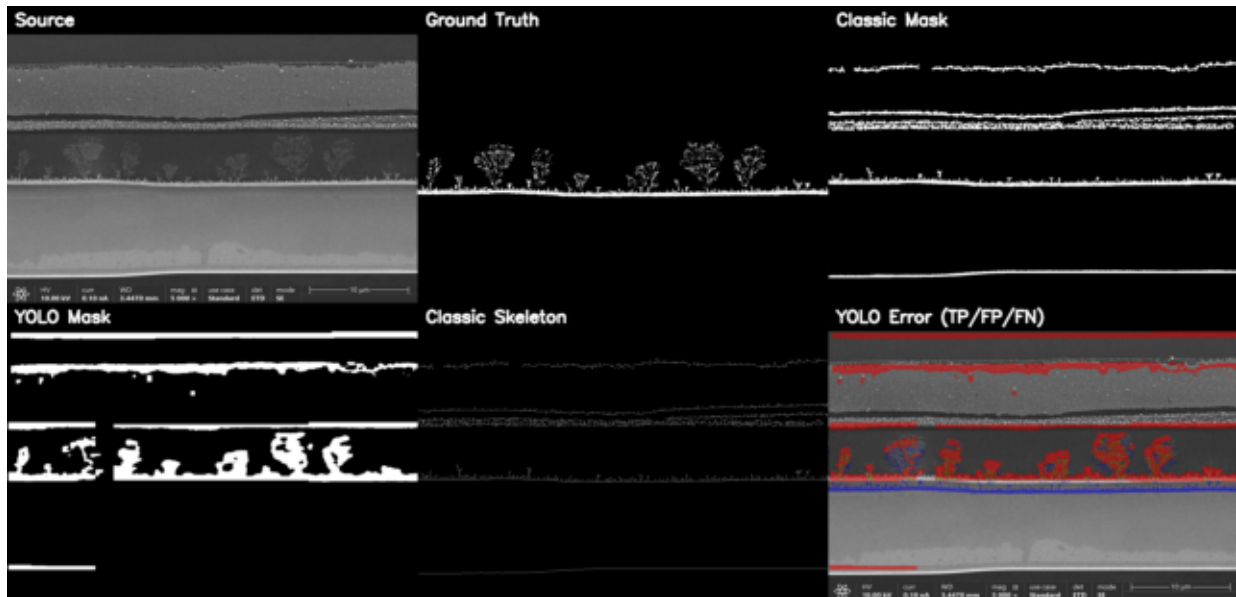
Success case: 2e-9_100s_014 (combined Dice=0.793)



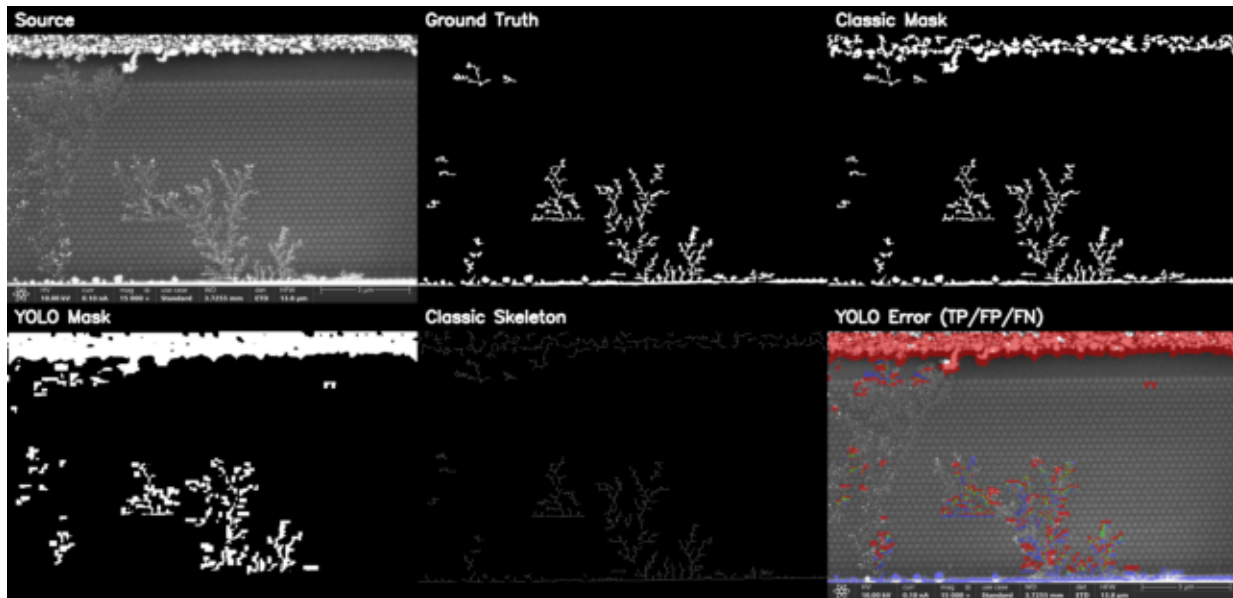
Success case: Ag_40nm_pitch_014 (combined Dice=0.757)



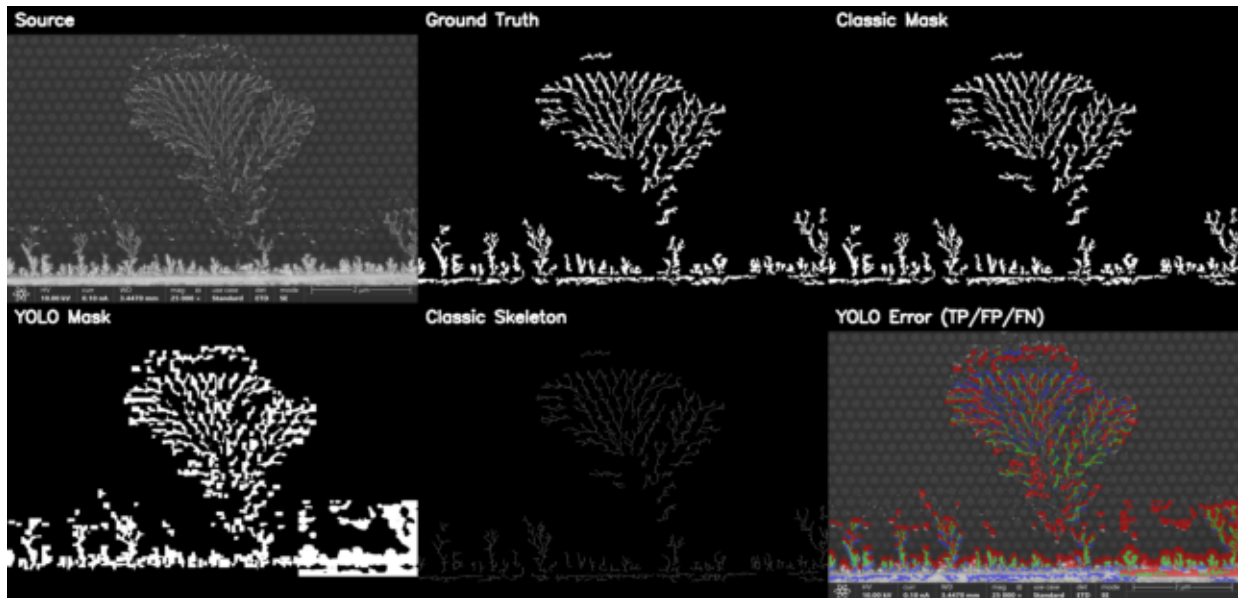
Failure case: 70nm_diameter_100nm_pitch_033 (combined Dice=0.044)



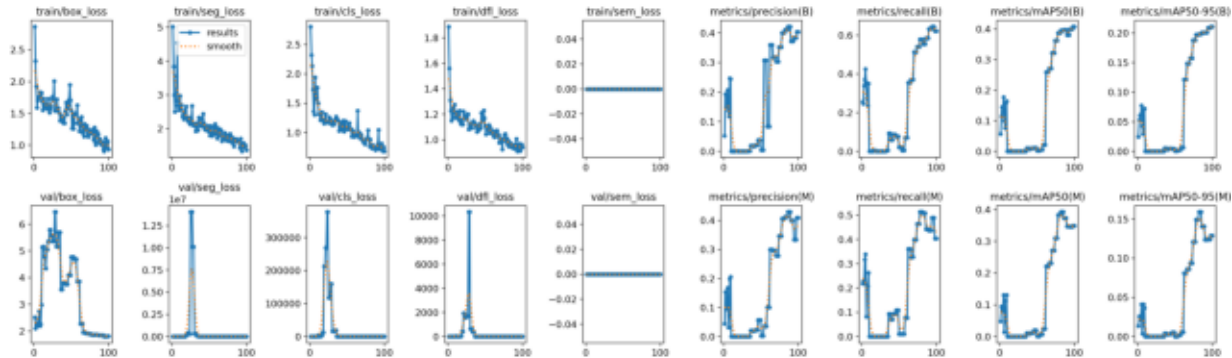
Failure case: 2e-9_100s_017 (combined Dice=0.401)



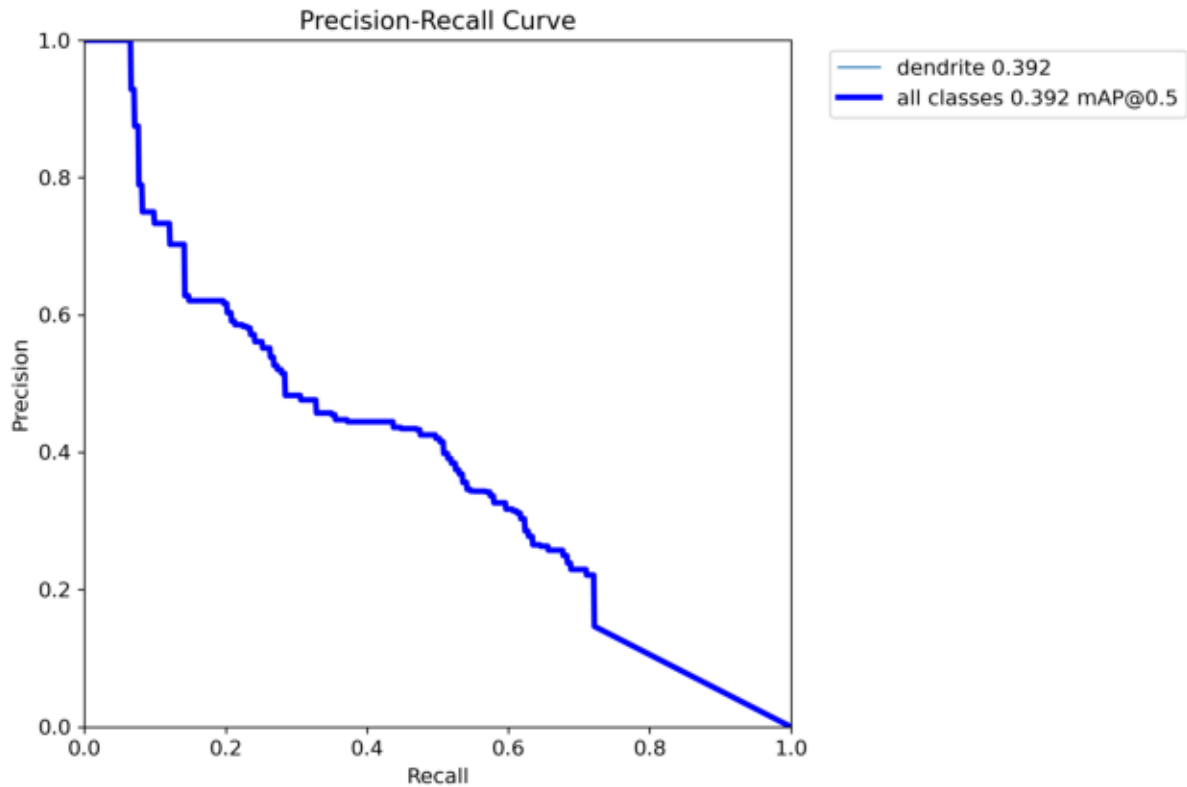
Failure case: 70nm_diameter_100nm_pitch_031 (combined Dice=0.679)



YOLO training metrics over epochs



Mask precision-recall curve



Normalized confusion matrix

